

Ejercicios, Libro Pösg. 44

$$\text{Dom}(f \circ g) = \{x \in \mathbb{R} : x \in \text{Dom}(g) \wedge g(x) \in \text{Dom}(f)\}$$

$$f(x) = \sqrt{x}, \quad g(x) = \sqrt{2-x}$$

$$\text{Dom } f(x) : \{x \in \mathbb{R} : \sqrt{x} \in \mathbb{R}\} = [0, +\infty[$$

$$\text{Dom } g(x) : \{x \in \mathbb{R} : \sqrt{2-x} \in \mathbb{R}\} = [-\infty, 2]$$

$$a) (f \circ g) : \{x \in \mathbb{R} : x \in [-\infty, 2] \wedge \sqrt{2-x} \in [0, +\infty[\}$$

$$\begin{aligned} f \circ g &= \sqrt{2-x} \geq 0 \quad ()^2 \\ 2-x &\geq 0 \\ x &\leq 2 \\ &[-\infty, 2] \end{aligned}$$

$$f \circ f = \{x \in \mathbb{R} : x \in [0, +\infty[\wedge \sqrt{x} \geq 0\}$$

$$\sqrt{x} \geq 0$$

$$[0, +\infty[\cap [-\infty, 2] = [0, 2]$$

$$\begin{aligned} [0, 2] &\rightarrow \mathbb{R} \\ f \circ g &= \sqrt{2-x} \end{aligned}$$

$$b) (g \circ f) : \{x \in [0, +\infty[\wedge \sqrt{x} \leq 2\}$$

$$\sqrt{x} \leq 2, \quad x \leq 4, \quad [-\infty, 4]$$

$$\begin{aligned} [0, +\infty[\cap [-\infty, 4] \\ [0, 4] \end{aligned}$$

$$\begin{aligned} [0, 4] &\rightarrow \mathbb{R} \\ g \circ f &= \sqrt{2-\sqrt{x}} \end{aligned}$$

$$1) f(x) = \sqrt{25-x^2}, \quad g(x) = \sqrt{x+1}$$

$$2) f(x) = \frac{1}{x-1}, \quad g(x) = \frac{1}{x} - 2$$

$$1) \text{Dom } f: \{x \in \mathbb{R} : \sqrt{25-x^2} \in \mathbb{R}\} : 25-x^2 \geq 0 \Rightarrow 5 \geq |x| \quad \left. \begin{array}{l} x \leq 5 \quad]-\infty, 5] \\ x \geq -5 \quad [-5, +\infty[\end{array} \right\} [-5, 5]$$

$$\text{Dom } g(x): \{x \in \mathbb{R} : \sqrt{x+1} \in \mathbb{R}\} : x+1 \geq 0 \Rightarrow x \geq -1 \quad - [-1, +\infty[$$

$$2) \text{Dom } f: \{x \in \mathbb{R} : \frac{1}{x-1} \in \mathbb{R}\} : \mathbb{R} - \{1\}$$

$$\text{Dom } g: \{x \in \mathbb{R} : \frac{1}{x} - 2 \in \mathbb{R}\} : \mathbb{R} - \{0\}$$

$f+g$)

$$1) [-5, 5] \cap [-1, +\infty[= [-1, 5] \rightarrow \mathbb{R}$$

$$f+g = \sqrt{25-x^2} + \sqrt{x+1}$$

$$2) \mathbb{R} - \{1\} \cap \mathbb{R} - \{0\} = \mathbb{R} - \{1, 0\} \rightarrow \mathbb{R}$$

$$f+g = \frac{-2x^2 + 4x - 1}{x(x-1)}$$

$f \cdot g$)

$$1) [-1, 5] \rightarrow \mathbb{R}$$

$$f \cdot g = \frac{\sqrt{25-x^2} \cdot \sqrt{x+1}}{\sqrt{-x^3-x^2+25x+25}}$$

$$2) \mathbb{R} - \{1, 0\} \rightarrow \mathbb{R}$$

$$f \cdot g = \frac{-2x+1}{x(x-1)}$$

$$f(x) = x^3 + 5, \quad g(x) = \sqrt[3]{x}$$

$$\text{Dom } f: \{x \in \mathbb{R} : x^3 + 5 \in \mathbb{R}\} : \mathbb{R}$$

$$\text{Dom } g(x): \{x \in \mathbb{R} : \sqrt[3]{x} \in \mathbb{R}\} : [0, +\infty[$$

$$f \circ g: \{x \in \mathbb{R} : x \in [0, +\infty[\wedge \sqrt[3]{x} \in \mathbb{R}\} \quad g \circ f: \{x \in \mathbb{R} : x \in \mathbb{R} \wedge x^3 + 5 \geq 0\}$$

$$\sqrt[3]{x} \in \underbrace{]-\infty, 0]}_{x: \text{Indefinit}} \cup [0, +\infty[$$

$$\sqrt[3]{x} \geq 0$$

$$[0, +\infty[$$

$$[0, +\infty[\rightarrow \mathbb{R}$$

$$f \circ g = x + 5$$

$$x^3 \geq -5$$

$$x \geq \sqrt[3]{-5}$$

$$x \geq \sqrt[3]{-5}$$

$$[-\sqrt[3]{5}, +\infty[\cap \mathbb{R}$$

$$[-\sqrt[3]{5}, +\infty[\rightarrow \mathbb{R}$$

$$g \circ f = \sqrt[3]{x^3 + 5}$$

$$a) \log_3 81$$

$$3^x = 81 \Rightarrow x = 4$$

$$b) \log_3 \left(\frac{1}{81} \right)$$

$$3^x = \frac{1}{81} \Rightarrow x = -4$$

$$c) \log_9 3 \Rightarrow 9^x = 3$$

$$3^{2x} = 3$$

$$2x = 1$$

$$x = \frac{1}{2}$$

$$d) \ln \left(\frac{1}{e^2} \right) = \ln(e^{-2}) = -2$$

$$e) \ln(\sqrt{e}) \Rightarrow \ln(e^{1/2}) = 1/2$$

$$f) \ln(\ln(e^{e^{50}})) = \ln(e^{50}) = 50$$

$$g) \log_2(30) - \log_2(15)$$

$$\log_2 \left(\frac{30}{15} \right) = \log_2(2) = 1$$

$$h) \log_3(10) - \log_3(5) - \log_3(18)$$

$$\log_3 \left(\frac{10}{5 \cdot 18} \right) = \log_3 \left(\frac{10}{90} \right) = \log_3 \left(\frac{1}{9} \right) = \log_3(3^{-2}) = -2$$

$$i) 2 \log_5(100) - 4 \log_5(50)$$

$$2(\log_5(100) - 2 \log_5(50))$$

$$2 \left(\log_5 \left(\frac{100}{50^2} \right) \right) \Rightarrow 2(\log_5(5^{-2})) \Rightarrow 2 \cdot -2 \Rightarrow -4$$